# ANDRY PAEZ

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## EDUCATION

**B.S., Physics** **May 2026**

*San Jose State University, San Jose, CA. GPA 3.80. Dean's Scholar (Fall 2024, Fall 2025)*

**A.S.-T, Computer Science; A.S.-T, Mathematics; A.S.-T, Physics; A.A., Language and Rationality** **May 2024**

*Modesto Junior College, Modesto, CA. GPA 3.73. Graduation With Honors; President's List (5 terms)*

**Relevant Coursework:** Quantum Mechanics, Quantum Information Science, Computational Methods in Physics, Advanced Mechanics, Electricity and Magnetism I/II, Thermodynamics and Statistical Physics, Mathematical Methods, Linear Algebra, Ordinary Differential Equations, Data Structures and Algorithms, UNIX/Linux Systems, Networking

## RESEARCH EXPERIENCE

**Research Assistant, Dept. of Physics and Astronomy** **Feb 2026 – Present**

*San Jose State University (advisor: Dr. Hilary Hurst), 30 hrs/week*

- Develop spinor Bose-Einstein condensate simulations (1D Gross-Pitaevskii equation): imaginary-time ground-state preparation, time-splitting spectral (Strang) evolution, and Bogoliubov-de Gennes spectra; all work merged into the research group's codebase through advisor-reviewed pull requests.
- Vectorized the sine-DVR kinetic-energy operator, achieving up to 1,880x speedup (161 s to 0.09 s at a 2048-point grid) with bit-exact numerical parity verified by a benchmark harness.
- Built the initial-condition generation pipeline: CLI with HDF5 output, hash-addressed run catalog with a rebuildable SQLite index, and a Slurm write-then-index workflow for the SJSU HPC cluster that avoids concurrent-writer contention.
- Designed controlled storage benchmarks (compression costs 55x write time for 11% disk savings) and authored the store-vs-recompute recommendation adopted as the project default.
- Validated spinor ground states to machine precision (norm error 2e-16) with correct miscible/immiscible domain-wall structure, and demonstrated the expected inverse-square-root scaling of domain-wall width with interaction strength.
- Implemented interaction-quench studies of spontaneous symmetry breaking with Truncated Wigner vacuum-noise seeding and domain-wall counting, extracting Kibble-Zurek defect scaling.
- Authored dated technical reports (benchmarks, validation, design recommendations) whose conclusions set project defaults.

## PROJECT EXPERIENCE

**Reachy Mini Robot Control Console** **Jun 2026 – Present**

- Built a FastAPI web control console for a Reachy Mini robot hosted on a headless Jetson Orin Nano, designed as a pure HTTP/WebSocket proxy so the robot daemon stays the single writer to the hardware; functionally verified against the live robot.
- Root-caused hardware/systems failures across the stack: compiled GStreamer webrtcsink from source for arm64 with NVMM buffer transport, traced post-OS-migration speaker silence to ALSA card remapping and sink volume, and fixed a signal-forwarding bug in process management with a process-tree signaller.
- Reduced idle UI overhead to zero DOM writes per tick via change-guarded rendering and SSE rate reduction, fixing a measured phone battery/heat problem; verified behavior by driving the served app with Playwright.
- Extended the platform with an in-place app library and Hermes voice agent, routing local Whisper STT through Hermes to neural TTS while supporting cancellation, queued voice follow-ups, access-controlled native-session reuse, and live camera sharing confirmed on the Jetson.

**Kalshi Weather-Market Trading System** **Jan 2026 – Mar 2026**

- Built an end-to-end quantitative trading system for weather prediction markets: NWS and commercial forecast ingestion, a Gaussian edge model, stacked LSTM temperature models (PyTorch) trained per city, a trading engine handling market bracket parsing and station rounding compensation, and a Streamlit monitoring dashboard.
- Documented methodology and results in a LaTeX technical report with supporting analysis notebooks.

**Sage, Local-First Knowledge Agent** **Feb 2025 – May 2026**

- Designed and built a desktop knowledge agent sustained over 15 months: document ingestion into a local SQLite store with embeddings, grounded chat with citations, and quiz/flashcard/report/audio generation.
- Stack: Tauri v2, Next.js 15/React 19, FastAPI, multi-provider LLM abstraction (OpenAI, Anthropic, Ollama, DeepSeek), pytest suite with GitHub Actions CI.

**LocalLM, Agentic Desktop App for Local Models** **Apr 2026**

- Built a cross-platform desktop app for local Ollama models with an agentic tool-calling layer (web search, file I/O, code execution) implemented in Rust (Tauri 2) with a React/TypeScript frontend, streaming chat, and a model recommendation registry with machine-spec detection for quantization-aware model selection.

**omnisense, Offline Voice and Vision Assistant** **Apr 2026**

- Built a modular multimodal assistant package for Jetson Orin Nano: camera abstraction with GStreamer/OpenCV/mock backends, faster-whisper speech-to-text, voice activity detection, YOLO overlay, and a FastAPI REST/WebSocket interface, testable without hardware via mock backends.

**Quantum Portfolio Optimization, Womanium and WISER Program** **Aug 2025**

- Implemented QAOA as a new ansatz family in an end-to-end portfolio optimization workflow using qiskit-algorithms with a custom local-search optimizer and Design-of-Experiments hyperparameter exposure.
- Benchmarked against the classical CPLEX solver on a 31-bond portfolio: median objective gap of 5.6% with QAOA alone, reduced to 0.02% after local search with about 20 additional objective evaluations.

**2D Ising Model Phase-Transition Study** **Apr 2026 – May 2026**

- Built a Metropolis Monte Carlo simulation recovering the critical temperature (Tc of about 2.269 J/kB), with local energy-change optimization and small-lattice validation; documented in a LaTeX report and presentation.

**Quantum Error Correction Using Shor's Code** **Dec 2024**

- Implemented Shor's 9-qubit and 3-bit repetition codes in Qiskit; simulated under controlled noise in Qiskit Aer and ran on IBM quantum hardware via Qiskit Runtime, measuring a 36% fidelity improvement with error correction.

**The Seedling Project** **Dec 2023 – Feb 2024**

- Led full-stack development of an educational web platform (React, Django, Wagtail headless CMS; 100+ resources across 15+ STEM courses) on a 20-member team; deployed at seedlingeducation.org. SolveCC Competition finalist.

## OPEN-SOURCE CONTRIBUTIONS

- **NousResearch/hermes-agent:** three merged bugfixes to a large open-source agent framework: a systemd gateway restart race, a Discord 8KB command-payload split, and Electron main-window lifecycle recovery after a closed-window second launch.
- **crafter-station/petdex:** merged fix for a sidecar child-process zombie leak (reap children on respawn) in a desktop app's process supervisor.
- **hhurst/Quantum_Gases:** packaging, CLI, and cross-platform contributions merged into a research group's simulation codebase (see Research Experience).

## ADDITIONAL EXPERIENCE

**Supplemental Instructor, Computer Science, MESA Program** **Jan 2024 – Jun 2024**

*Modesto Junior College, Modesto, CA*

- Provided weekly tutoring to 5-6 students in data structures, programming fundamentals, Linux/VM troubleshooting, and Vim.

## CERTIFICATES

2025 Quantum Program, WISER (Aug 2025); Quantum Algorithms for Portfolio Optimization, WISER (Aug 2025); Quantum Algorithms for PDEs (Module 2) and Nonlinear Problems (Module 5), WISER (Aug 2025); QBronze169 Diploma, QWorld (Aug 2025); PennyLane LCU Challenge, Womanium and WISER (Jul 2025)

## TECHNICAL SKILLS

- **Languages:** Python (primary), C/C++, TypeScript/JavaScript, Rust, SQL, Bash, LaTeX
- **Scientific Computing:** NumPy, spectral/FFT methods, Monte Carlo, PyTorch, Qiskit, HDF5, Slurm/HPC, Jupyter, Matplotlib
- **AI Systems:** local LLM inference (llama.cpp, Ollama), agent architectures and tool-calling loops, RAG, MCP, edge deployment (Jetson), faster-whisper; multi-agent coding orchestration with verification-first review gates
- **LLM Internals:** quantization formats (K-quants, importance-matrix I-quants) and quantization-aware model selection, LoRA fine-tuning, KV-cache memory/compression tradeoffs, MoE expert pruning
- **Software:** FastAPI, Next.js/React, Tauri, WebSockets/SSE, pytest, GitHub Actions CI, Docker, Playwright, Linux systems (systemd, GStreamer, ALSA/PipeWire), Git
- **Hardware and Lab:** Jetson Orin Nano, ESP32 firmware (PlatformIO/FreeRTOS), oscilloscopes, DMMs, spectrum analyzers

## ACHIEVEMENTS

- SolveCC Competition Finalist, Modesto Junior College, Feb 2024
- 2nd Annual Integration Bee Finalist, Modesto Junior College, Mar 2024